

Summary of DCASE 2026 Task 5: Audio-Dependent Question Answering

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Abstract—DCASE 2026 Task 5 introduces Audio-Dependent Question Answering (ADQA), which tests whether large audio-language models answer from the audio rather than from textual priors. An Audio-Dependency Filtering (ADF) pipeline combines silent-audio probing, perplexity, a large language model (LLM) commonsense check, and human review to remove items solvable from text alone. The 3000 items that pass form the ADQA-Bench evaluation set, spanning music, speech, and environmental audio. The inaugural edition draws 14 teams and 36 submissions across two tracks defined by total parameter count (up to 100B and under 10B). A Chung-Ang University ensemble of MOSS-Audio-8B-Thinking (MOSS-8B-Think) and Qwen3-Omni-30B reaches the top overall accuracy at 58.33 %, and a MOSS-8B-Think configuration from the same team leads the sub-10B track at 57.30 %. Across the 30 submissions with a comparable development score, evaluation accuracy falls by 11.91 percentage points (pp) on average (median 10.91 pp) on the hidden evaluation split. The most common building blocks are: the MOSS-8B-Think backbone (13 of 36 submissions), Low-Rank Adaptation (LoRA) fine-tuning on AudioMCQ-StrongAC, and preference or reinforcement-learning objectives—Group Relative Policy Optimization (GRPO) in five teams, Group reward-Decoupled Normalization Policy Optimization (GDPO) in two. At test time, prompt engineering is near-universal, and majority or choice-permutation voting is common. Every system misses the same set of 233 evaluation items.

Index Terms—Audio question answering, large audio-language models, multimodal evaluation, DCASE challenge

1. INTRODUCTION

Large Audio-Language Models (LALMs) can inflate their accuracy on multiple-choice audio benchmarks by exploiting textual shortcuts. When the audio is replaced with silence, state-of-the-art LALMs can still answer many questions correctly, using linguistic priors and dataset artifacts [1]–[4]. Progress requires benchmarks and models that depend on the audio signal.

DCASE 2026 Task 5, Audio-Dependent Question Answering (ADQA) [1], addresses this gap. An Audio-Dependency Filtering (ADF) pipeline drops questions that a text-only model can solve, yielding a 3000-item evaluation set (ADQA-Bench). Submissions are open-source LALMs that answer audio-grounded multiple-choice questions, usually with four options. Two tracks use total parameter count: an Overall track ($\leq 100B$) and a Lightweight track ($< 10B$).

The top Overall system, Lim_CAU from Chung-Ang University [5], reaches 58.33 %; the same team wins the Lightweight track at 57.30 %. The baselines are evaluated only on the development set. The strongest organizer baseline, Qwen3-Omni-30B-A3B-Instruct (hereafter Qwen3-Omni-30B) [6], scores 62.48 %. Team submissions average ~ 12 pp below their own development scores on the hidden split [1] (Section 5.2); baseline and team evaluation scores thus refer to different splits. Every system exceeds the random-guess reference (25.46 %; nominal four-option chance 25 %). The median across the 36

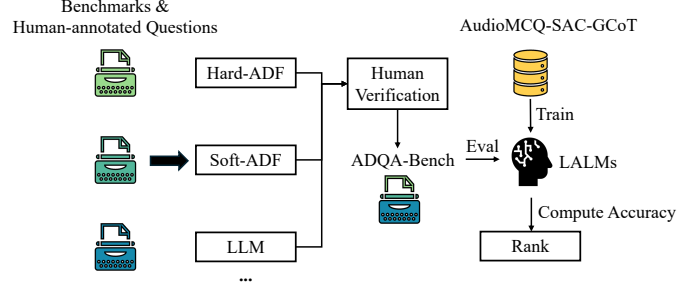


Fig. 1: Audio-Dependency Filtering (ADF). Three filtering stages (Hard-ADF, Soft-ADF, large language model (LLM) commonsense) and a final human verification step yield ADQA-Bench from existing benchmarks and newly annotated multiple-choice-question (MCQ) items; training items follow the related Strong Audio-Contribution (StrongAC) selection on the AudioMCQ corpus [1], with chain-of-thought (CoT) rationales; the figure’s AudioMCQ-SAC-GCoT block denotes this AudioMCQ-StrongAC-GeminiCoT training corpus (SAC = StrongAC; GCoT = Gemini-distilled CoT).

submissions is 50.87 %, and the top score of 58.33 % leaves headroom on ADQA-Bench.

Section 2 details the task and ADF, Section 3 the submissions, Section 4 the methods, Section 5 the performance analysis, and Section 6 the discussion and future directions.

2. TASK SETUP AND DATASET

2.1. Task Definition

DCASE 2026 Task 5 (ADQA) is a multiple-choice question (MCQ) task over audio clips. Each item pairs one recording with a question and answer options, usually four, with a single correct choice. Questions must be *audio-dependent*: text alone is insufficient.

2.2. Audio-Dependency Filtering (ADF)

The ADF cascade [1] is a four-stage pipeline (Fig. 1) that keeps only items whose correct answer depends on the audio signal.

Hard-ADF (silent-audio probe). We replace the audio with silence and ask a pool of LALMs to answer. An item is discarded if more than one blind model still picks the correct option, since the answer is then largely recoverable without listening.

Soft-ADF (perplexity filter). We score each option with language-model perplexity under silent audio. If the correct option has the lowest perplexity, the item is dropped: a text-only preference for the right answer exposes a linguistic shortcut.

LLM commonsense filter. A text-only large language model (LLM) sees the question and options without the audio. Items it answers correctly from commonsense or world knowledge are pruned.

Human verification. A two-pass human review checks that the labeled answer is correct given the audio and that every distractor is plausible but acoustically wrong.

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Table 1: Top-1 accuracy of organizer baselines on the development set. Random-guess level: 25.46 %.

Baseline system	Accuracy
Fun-Audio-Chat-8B [8]	56.81 %
Kimi-Audio-7B [9]	46.36 %
MiMo-Audio-7B [10]	54.57 %
Qwen3-Omni-30B [6]	62.48 %
Step-Audio-2 Mini [11]	50.53 %
Random guess	25.46 %

2.3. Datasets

Training set. The official training corpus, AudioMCQ-StrongAC-GeminiCoT (hereafter AudioMCQ-StrongAC), is derived from the ~571k-item AudioMCQ corpus [1]. We keep a subset of the Strong Audio-Contribution (StrongAC) split and add chain-of-thought (CoT) rationales distilled from Gemini 3.1 Pro.

Development set. 1607 MCQ items drawn from existing benchmarks (MMAU [2], MMAR [3], MMSU [4]) together with newly annotated questions, all passing the four-stage ADF.

Evaluation set. The hidden ADQA-Bench contains 3,000 MCQ items filtered under the same protocol. Team scores refer to evaluation; organizer baselines refer to development (Section 5.2).

Protocol. The Overall track allows $\leq 100B$ total parameters, with no component $> 30B$; Lightweight requires $< 10B$. Each team can submit up to four open-source LALM systems. Evaluation answers remain hidden during submission. Teams cannot manually label evaluation audio, re-transcribe it for training, or feed human answers back into their systems.

Leakage safeguards and limits. AudioMCQ uses source-dataset training partitions [1]; public development data supports system selection. Withheld evaluation answers and the annotation ban limit direct label leakage, but do not certify absence of benchmark overlap in pretrained backbones or external training data.

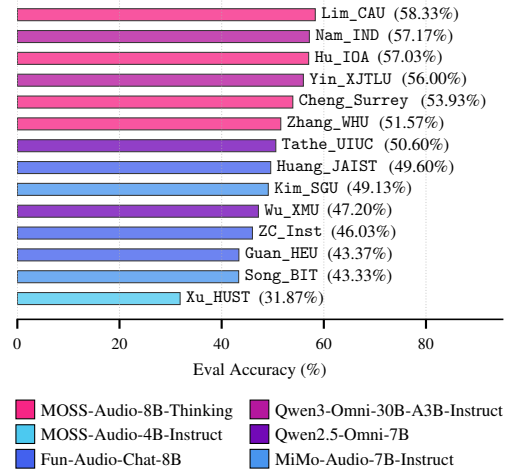
2.4. Baseline Systems

The organizers release five LALM baselines with tuned prompts, evaluated by development-set top-1 accuracy (Table 1). The random-guess level is 25.46 %: averaging $1/K_i$ over items with K_i options gives a value above the four-option 25 %, because option counts vary [7]. Qwen3-Omni-30B scores highest.

3. SUBMISSION OVERVIEW

The first edition of DCASE 2026 Task 5 draws 14 teams and 36 ranked submissions, spanning 31.87 % to 58.33 % accuracy. Because each system is scored on 3000 items, gaps of about a percentage point warrant paired uncertainty analysis. Systems use six public LALMs and differ in backbone, data curation, and inference strategy.

The overall champion is **Lim_CAU_4** from Chung-Ang University [5], which reaches 58.33 % by ensembling MOSS-Audio-8B-Thinking (MOSS-8B-Think) [12] with Qwen3-Omni-30B [6] (96.0 B in the challenge metadata; the team reports a 92 B effective total over four MOSS-8B-Think and two Qwen3-Omni-30B inference views). Its pipeline chains audio-vs-silent difficulty binning, Group reward-Decoupled Normalization Policy Optimization (GDPO) [13]-based Low-Rank Adaptation (LoRA) [14], an inference-time acoustic tagger, and choice-permutation ensembling (see Section 4.4). **Nam_IND** [15] applies 4-bit quantized LoRA (QLoRA) [16] supervised fine-tuning (SFT) with a response normalizer on Qwen3-Omni-30B (60 B, 57.17 %). **Hu_IOA** [17] runs MOSS-8B-Think with zero-shot label-only decoding and 5-view choice-permutation majority voting (57.03 %).

**Fig. 2:** Best evaluation accuracy per team, colored by primary large audio-language model (LALM). The dashed line marks the 49.65 % mean across the 14 teams’ best submissions; four teams reach at least 56 %.

Yin_XJTLU [18] builds a training-free Qwen3-Omni-30B prompt-program ensemble (56.00 %). **Cheng_Surrey** [19] adds retrieval augmentation to MOSS-8B-Think (53.93 %), and **Zhang_WHU** [20] combines SFT with reinforcement learning (RL) on the same backbone (51.57 %). The remaining eight teams [21]–[28] span from 31.87 % to 50.60 %.

Table 2 lists all 36 ranked submissions and marks lightweight systems ($< 10B$) with †; Fig. 2 plots each team’s best accuracy. The lightweight champion is **Lim_CAU_1** at 57.30 %, 1.03 pp behind the 96 B ensemble.

MOSS-8B-Think leads backbone adoption among top systems. Four of 14 teams (**Lim_CAU**, **Hu_IOA**, **Cheng_Surrey**, **Zhang_WHU**) pick it as the backbone of their best submission; **Yin_XJTLU** uses it in three of four submissions. **Hu_IOA**’s untuned MOSS-8B-Think reaches 57.03 %, versus 49.87 % for its tuned Qwen2.5-Omni-7B [17], [29]; this comparison confounds backbone, size, and tuning. Controlled ablations are needed to separate checkpoint alignment, LoRA [14] compatibility (**Lim_CAU**, **Zhang_WHU**), and retrieval support (**Cheng_Surrey**).

Among the top five teams, **Yin_XJTLU**’s Qwen3-Omni prompt-program ensemble (**Yin_XJTLU_1**, 56.00 %) narrowly beats its MOSS-8B-Think variants: audio-token attention biases (**Yin_XJTLU_2**/**Yin_XJTLU_3**) and an option-elimination prompt program (**Yin_XJTLU_4**).

4. METHODOLOGIES AND TRENDS

We compare the 36 submissions by backbone and technique, and the 14 teams by training paradigm—one label per team, taken from its best system and applied to all of that team’s submissions. Paradigm comparisons are descriptive only: cell sizes (1–5 teams) are too small for statistical inference, and paradigm is closely confounded with each team’s backbone choice.

4.1. Paradigm Taxonomy

Each team’s best system falls into one of six paradigms (see Table 2), reported as best, mean, and team count (best omitted where not informative): hybrid (58.33 %, 54.83 %, 3), training-free (56.00 %, 51.60 %, 2), supervised fine-tuning with reinforcement learning (SFT+RL; mean 51.09 %, 2 teams), SFT only (mean 45.61 %, 5 teams), and RL only and retrieval-augmented (1 team each). **Hybrid** systems fuse components from different paradigms [5], [17], [23]

Table 2: All 36 ranked DCASE 2026 Task 5 submissions, sorted by evaluation accuracy. Bold marks the overall champion; † marks lightweight systems (< 10B total parameters). Dev/Eval = development/evaluation top-1 accuracy; Diff = Eval – Dev in pp; – = no published development score. Paradigm labels follow each team’s best system and apply to all its submissions: Hybrid (H), supervised fine-tuning (SFT; S), SFT with reinforcement learning (RL; S+R), RL only (RL), Training-free (TF), Retrieval-augmented generation (RAG). Base LALM = large audio-language model. Model names: MOSS-Audio-8B-Thinking (MOSS-8B-Think); Qwen3-Omni-30B-A3B-Instruct (Qwen3-Omni-30B); Qwen3-Embedding-0.6B (Qwen3-Emb); MiMo-Audio-7B-Instruct (MiMo-Audio-7B); MOSS-Audio-4B-Instruct (MOSS-4B-Instruct). The 80.0B for Lim_CAU_3 follows the metadata’s replicated-view convention: one 8B MOSS-8B-Think checkpoint across ten permutation views (8B unique weights). The 96.0B for Lim_CAU_4 also follows metadata; its report gives a 92B effective total over six inference views. °: Yin_XJTLU’s category-aware audio-attention oracle upper bound; deployed-system calibrated dev accuracy is not separately reported.

Rank	Submission	Institution	Paradigm	Base LALM	Params	Dev (%)	Eval (%)	Diff (pp)
1	Lim_CAU_4 [5]	Chung-Ang U.	H	MOSS-8B-Think + Qwen3-Omni-30B	96.0B	70.50	58.33	–12.17
2	Lim_CAU_3 [5]	Chung-Ang U.	H	MOSS-8B-Think	80.0B	70.01	58.10	–11.91
3	Lim_CAU_1 [5]†	Chung-Ang U.	H	MOSS-8B-Think	8.0B	69.63	57.30	–12.33
4	Nam_IND_2 [15]	Independent	S	Qwen3-Omni-30B	60.0B	–	57.17	–
5	Nam_IND_4 [15]	Independent	S	Qwen3-Omni-30B + Gemma-4-E4B-it	94.0B	–	57.13	–
6	Lim_CAU_2 [5]†	Chung-Ang U.	H	MOSS-8B-Think	8.0B	69.70	57.07	–12.63
7	Hu_IOA_4 [17]†	IOA, CAS	H	MOSS-8B-Think + Qwen3-Emb	8.6B	67.70	57.03	–10.67
8	Nam_IND_3 [15]	Independent	S	Qwen3-Omni-30B	60.0B	–	56.73	–
9	Hu_IOA_3 [17]†	IOA, CAS	H	MOSS-8B-Think + Qwen3-Emb	8.6B	66.02	56.70	–9.32
10	Yin_XJTLU_1 [18]	XJTLU	TF	Qwen3-Omni-30B	30.0B	68.33	56.00	–12.33
11	Nam_IND_1 [15]	Independent	S	Qwen3-Omni-30B	60.0B	67.27	55.90	–11.37
12	Yin_XJTLU_4 [18]†	XJTLU	TF	MOSS-8B-Think	8.0B	66.40	55.80	–10.60
13	Yin_XJTLU_2 [18]†	XJTLU	TF	MOSS-8B-Think	8.0B	64.97	55.60	–9.37
14	Yin_XJTLU_3 [18]†	XJTLU	TF	MOSS-8B-Think	8.0B	66.46°	55.27	–11.19
15	Cheng_Surrey_1 [19]†	Surrey	RAG	MOSS-8B-Think	8.0B	65.07	53.93	–11.14
16	Cheng_Surrey_2 [19]†	Surrey	RAG	MOSS-8B-Think	8.0B	65.01	53.50	–11.51
17	Zhang_WHU_1 [20]†	WHU + CUHK-SZ	S+R	MOSS-8B-Think	8.0B	62.79	51.57	–11.22
18	Zhang_WHU_2 [20]†	WHU + CUHK-SZ	S+R	MOSS-8B-Think	8.0B	62.79	51.13	–11.66
19	Tathe_UIUC_1 [21]†	UIUC + CMU	S+R	Qwen2.5-Omni-7B	7.0B	58.43	50.60	–7.83
20	Hu_IOA_2 [17]†	IOA, CAS	H	Qwen2.5-Omni-7B + Qwen3-Emb	7.6B	58.93	49.87	–9.06
21	Tathe_UIUC_2 [21]†	UIUC + CMU	S+R	Qwen2.5-Omni-7B	7.0B	57.31	49.83	–7.48
22	Hu_IOA_1 [17]†	IOA, CAS	H	Qwen2.5-Omni-7B + Qwen3-Emb	7.6B	58.93	49.63	–9.30
23	Huang_JAIST_1 [22]†	JAIST	S	Fun-Audio-Chat-8B	8.0B	64.84	49.60	–15.24
24	Kim_SGU_4 [23]†	Sogang U.	H	MiMo-Audio-7B	7.0B	–	49.13	–
25	Kim_SGU_3 [23]†	Sogang U.	H	MiMo-Audio-7B	7.0B	59.43	48.87	–10.56
26	Tathe_UIUC_3 [21]†	UIUC + CMU	S+R	Qwen2.5-Omni-7B	7.0B	56.32	48.67	–7.65
27	Tathe_UIUC_4 [21]†	UIUC + CMU	S+R	Qwen2.5-Omni-7B	7.0B	55.51	48.33	–7.18
28	Kim_SGU_2 [23]†	Sogang U.	H	MiMo-Audio-7B	7.0B	73.43	47.97	–25.46
29	Wu_XMU_1 [24]†	Xiamen + THU	TF	Qwen2.5-Omni-7B	7.0B	53.90	47.20	–6.70
30	Kim_SGU_1 [23]†	Sogang U.	H	MiMo-Audio-7B	7.0B	72.76	46.77	–25.99
31	ZC_Inst_1 [25]†	HDU	S	Fun-Audio-Chat-8B	8.0B	64.45	46.03	–18.42
32	Guan_HEU_1 [26]†	HEU + UTS	S	Fun-Audio-Chat-8B	8.0B	53.08	43.37	–9.71
33	Song_BIT_1 [27]†	BIT	RL	MiMo-Audio-7B	7.0B	50.53	43.33	–7.20
34	Song_BIT_2 [27]†	BIT	RL	MiMo-Audio-7B	7.0B	48.10	41.53	–6.57
35	Song_BIT_3 [27]†	BIT	RL	MiMo-Audio-7B	7.0B	54.20	40.97	–13.23
36	Xu_HUST_1 [28]†	HUST	S	MOSS-4B-Instruct + DeBERTa-v3	5.4B	56.69	31.87	–24.82

and achieve the highest best accuracy and mean. Strong SFT-only (Nam_IND) and training-free (Yin_XJTLU) systems still land in the 55%–57% band, within 2.4 pp of the top hybrid system.

4.2. Base Model Choice

Table 3 groups submissions by primary backbone. Two models dominate. MOSS-8B-Think [12] (13 submissions, 5 teams) gives the best single-submission accuracy (58.33%, from a system that also fuses Qwen3-Omni-30B views) and a backbone mean of 55.49%. Qwen3-Omni-30B [6] (5 submissions, 2 teams) has a slightly higher raw mean (56.59%), but four of its five submissions come from one team’s SFT variants, so the cross-team mean is not meaningful at 2 teams. All other backbones plateau below 51% [21], [22], [24]–[28]. In the one available within-team comparison, added system size gives only marginal gains once a strong 8B backbone is in place: Lim_CAU’s 96B ensemble beats its lightweight entry by only 1.03 pp [5]. A single comparison cannot isolate the effect of size.

4.3. Commonly Used Techniques

Across the 36 submissions, most teams share a small pool of techniques on a common backbone. Table 4 lists the 16 technique tags used by at least two teams. Prompt engineering (11 teams), LoRA [14] (9), SFT (9), and CoT prompting (9) are the most common; Group Relative Policy Optimization (GRPO) [30] appears in 5 teams and majority voting in seven teams.

Table 3: Submissions grouped by primary large audio-language model (LALM; first model in submission metadata), sorted by best evaluation accuracy. Model names: MOSS-Audio-8B-Thinking (MOSS-8B-Think); Qwen3-Omni-30B-A3B-Instruct (Qwen3-Omni-30B); MOSS-Audio-4B-Instruct (MOSS-4B-Instruct); MiMo-Audio-7B-Instruct (MiMo-Audio-7B).

Base Model	#Sub.	#Teams	Best (%)	Mean (%)
MOSS-8B-Think	13	5	58.33	55.49
Qwen3-Omni-30B	5	2	57.17	56.59
Qwen2.5-Omni-7B	7	3	50.60	49.16
Fun-Audio-Chat-8B	3	3	49.60	46.33
MiMo-Audio-7B	7	2	49.13	45.51
MOSS-4B-Instruct	1	1	31.87	31.87

Choice-permutation ensembling: Lim_CAU [5] fuses MOSS-8B-Think and Qwen3-Omni views via weighted log-probs ($\lambda \approx 0.46$); Yin_XJTLU [18] votes across five prompt programs; Zhang_WHU [20] gains ~ 3.8 pp from eight-sample voting atop GRPO [30]. *RL objectives:* five teams train with GRPO [17], [20], [21], [23], [27], and Lim_CAU applies GDPO-style reward normalization on LoRA adapters [5]. *Reward engineering:* atop RL, Hu_IOA combines accuracy, format, question-type, embedding-similarity, and length-regularization rewards, optimized with GDPO [17], while Song_BIT adds an attention-anchoring reward that ties deep-layer cross-attention to audio key positions [27]. *Acoustic-feature injection:* Lim_CAU’s router [5] matches question keywords to categories—speech, speaker,

Table 4: Methodology tags adopted by at least two teams (16 tags total). “Top team” is the best-ranked team that uses the tag in any of its submissions, not necessarily in that team’s best-scoring one, and is not a performance attribution for the tag. Abbreviations: LoRA = Low-Rank Adaptation; SFT = supervised fine-tuning; GRPO = Group Relative Policy Optimization; GDPO = Group reward-Decoupled Normalization Policy Optimization.

Tag	#T	Top team	Tag	#T	Top team
Prompt Engineering	11	Lim_CAU	Self-Consistency	3	Yin_XJTLU
Chain-of-Thought	9	Hu_IOA	Voting Ensemble	3	Lim_CAU
LoRA	9	Lim_CAU	Choice-Permut. Ens.	3	Lim_CAU
SFT	9	Nam_IND	Reward Model	3	Zhang_WHU
Majority Voting	7	Lim_CAU	Knowledge Distill.	2	Zhang_WHU
GRPO	5	Hu_IOA	GDPO	2	Lim_CAU
Hard-Example Mining	4	Lim_CAU	LoRA (4-bit)	2	Nam_IND
Test-Time Scaling	4	Lim_CAU	Prompt Calibration	2	Lim_CAU

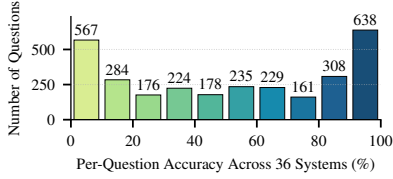


Fig. 3: Per-question accuracy across the 36 submissions, grouped into ten 10-percentage-point bins. Mass concentrates in the left-most (hardest) and right-most (easiest) bins.

music/scene—and runs only the relevant taggers: voice activity detection (VAD), Whisper [31] prosody, pyannote diarization [32], ECAPA-TDNN [33] speaker verification, and Contrastive Language-Audio Pretraining (CLAP) [34] scene matching, injecting the outputs as natural-language snippets. *SFT on AudioMCQ-StrongAC*: the dominant training recipe [17], [20]–[22], [26], with targets from answer text to structured or teacher-rewritten CoT rationales.

4.4. Winning System: Lim_CAU

The winning recipe [5] bins training candidates into six difficulty classes using matched (audio, silent) prompts. From 167,541 AudioMCQ-StrongAC [1] candidates, 12,800 rows are drawn with group-level quotas. Option D oversampling (A/B/C/D = 1.0/1.0/1.0/1.3) counters reported positional bias in MOSS-8B-Think outputs. GDPO [13]-based LoRA [14] then tunes the model. At inference, acoustic taggers inject librosa [35]-derived evidence. Lim_CAU_4 fuses MOSS-8B-Think and Qwen3-Omni-30B views via weighted log-probs ($\lambda \approx 0.46$) and reaches 58.33 %, the best overall accuracy. No ablation isolates the taggers’ gain from the ensemble.

5. PERFORMANCE ANALYSIS

5.1. Question-level Difficulty Distribution

Figure 3 bins the 3000 evaluation questions by the fraction of submissions that answer correctly. The distribution is bimodal: 567 questions (18.9 %) are solved by fewer than 10 % of systems and 638 (21.3 %) by more than 90 %. 233 items are missed by every ranked system; 110 are solved by all. For 47.6 % of items, fewer than half of the 36 submissions answer correctly, extending shared errors beyond the 233 universal failures (Section 6).

Two evaluation examples illustrate the listening demands: eval_0002 uses a speaker’s pause to distinguish seeking a German person from seeking a German-language teacher; eval_0120 asks what follows a refrigerator door closing (an electronic hum) [36]. These illustrate prosodic disambiguation and event ordering; error rates for these specific examples are not reported.

5.2. Dev-to-Eval Accuracy Drop

Audio-dependent filtering motivates a demanding evaluation split [1]. Of the 30 submissions with a comparable development score (excluding Nam_IND_2/3/4 and Kim_SGU_4, which publish none, and Song_BIT_3 and Yin_XJTLU_3, whose reported numbers are an MMAU test-mini proxy and an oracle upper bound), every one is lower on eval: mean gap -11.91 pp, median -10.91 pp, standard deviation (SD) 5.24 pp, ranging from -6.6 pp to -25.99 pp. Submissions share backbones and training data, so $n=30$ does not represent independent replicates; the SD is descriptive. Weighting the 14 teams equally gives a mean drop of 12.62 pp. The four comparable training-free submissions still lose 9.75 pp on average, showing that a gap persists without fine-tuning. By primary backbone, mean drops are 11.21 pp for MOSS-8B-Think (12 submissions), 7.89 pp for Qwen2.5-Omni-7B (7), and 15.16 pp for MiMo-Audio-7B (5).

The two teams with the largest gaps are Kim_SGU at -25.99 pp [23] and Xu_HUST at -24.82 pp [28]. The smallest gaps are Song_BIT at -6.6 pp/ -7.2 pp [27], Wu_XMU at -6.7 pp [24], and Tathe_UIUC at -7.2 pp to -7.83 pp [21], all with lower development scores. These comparisons cannot separate intrinsic item difficulty, source overlap, and development-set selection; even training-free systems can tune prompts. A controlled source- and question-type-matched comparison is needed to explain the gap causally.

5.3. Reasoning-Time Interventions

CoT-target SFT raises development accuracy for some systems—Hu_IOA’s Qwen2.5-Omni from 55.88 % to 57.93 % [17] and Zhang_WHU’s MOSS-8B-Think from 52.21 % to 56.94 % [20]—but three other reasoning-time add-ons do not. A separate training-time case shows the reverse risk: Hu_IOA reports a -8.71 pp development-split drop from SFT on MOSS-8B-Think, with development accuracy falling from 67.70 % to 58.99 %; they attribute the drop to catastrophic forgetting of the model’s native audio reasoning rather than to CoT supervision itself [17]. Nam_IND loses -1.93 pp with a CoT-then-letter prompt [15]. Tathe_UIUC’s Prometheus [37] CoT judge is statistically tied with its no-judge run (57.31 % vs 58.43 %, McNemar $p=0.23$) [21]. Yin_XJTLU drops -8.96 pp by replacing majority voting with a rule that picks the lowest-entropy branch across its training-free prompt programs [18]. This is a decoding-time change only, so it isolates the selection rule from CoT training.

6. DISCUSSION AND FUTURE DIRECTIONS

ADQA-Bench quantifies audio dependence: accuracy spans 31.87 %–58.33 %, with 233 items (7.8 %) missed by every system (Section 5.1). An oracle over the 36 submissions solves 2767 of 3000 items (92.2 %), well above every system. AudioMCQ-StrongAC derivatives [1], [17], [20], [21] dominate fine-tuning. CoT-target SFT helps some backbones, but CoT prompting, CoT-based judging, and entropy-based branch selection do not (Section 5.3). Acoustic grounding merits study (Section 4), though its gains remain unisolated [5], [27].

The dev-to-eval variance (Section 5.2) motivates a separate tuning split and overlap audits, as development reuses MMAU [2], MMAR [3] and MMSU [4] items (Section 2); tuning and source reuse remain confounded. The 233 universal failures (Section 5.1) need label and reasoning checks before attribution to encoders or pre-training. The 1.03 pp gap between the 96 B champion and the 8 B lightweight winner suggests that data curation and inference-time orchestration can rival raw scale. Open-ended generation can reduce elimination shortcuts but needs an audio-blind judge.

REFERENCES

- [1] H. He, X. Du, R. Sun, Z. Dai, Y. Xiao, M. Yang, J. Zhou, X. Li, Z. Liu, Z. Liang, C. Wu, Q. He, T. Lee, X. Chen, W.-L. Zheng, W. Wang, M. D. Plumbley, J. Liu, and Q. Kong, “Measuring audio’s impact on correctness: Audio-contribution-aware post-training of large audio language models,” arXiv preprint arXiv:2509.21060, 2025. [Online]. Available: <https://arxiv.org/abs/2509.21060>
- [2] S. Sakshi, U. Tyagi, S. Kumar, A. Seth, R. Selvakumar, O. Nieto, R. Duraiswami, S. Ghosh, and D. Manocha, “MMAU: A massive multi-task audio understanding and reasoning benchmark,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025. [Online]. Available: <https://arxiv.org/abs/2410.19168>
- [3] Z. Ma, Y. Ma, Y. Zhu, C. Yang, Y.-W. Chao, R. Xu, W. Chen, Y. Chen, Z. Chen, J. Cong, K. Li, K. Li, S. Li, X. Li, X. Li, Z. Lian, Y. Liang, M. Liu, Z. Niu, T. Wang, Y. Wang, Y. Wang, Y. Wu, G. Yang, J. Yu, R. Yuan, Z. Zheng, Z. Zhou, H. Zhu, W. Xue, E. Benetos, K. Yu, E.-S. Chng, and X. Chen, “MMAR: A challenging benchmark for deep reasoning in speech, audio, music, and their mix,” arXiv preprint arXiv:2505.13032, 2025. [Online]. Available: <https://arxiv.org/abs/2505.13032>
- [4] D. Wang, J. Li, J. Wu, D. Yang, X. Chen, T. Zhang, and H. Meng, “MMSU: A massive multi-task spoken language understanding and reasoning benchmark,” arXiv preprint arXiv:2506.04779, 2025. [Online]. Available: <https://arxiv.org/abs/2506.04779>
- [5] H. J. Kim, B. Kim, J. C. Ryu, Y. R. Kim, Y. Oh, B. E. Choi, C. Lim, and I.-Y. Kwak, “Audio-grounded hard-example training with acoustic tagging for audio-dependent question answering,” Chung-Ang Univ., DCASE2026 Challenge, Jun. 2026.
- [6] J. Xu *et al.*, “Qwen3-Omni technical report,” arXiv preprint arXiv:2509.17765, 2025. [Online]. Available: <https://arxiv.org/abs/2509.17765>
- [7] DCASE Task 5 Organizers, “DCASE 2026 Task 5 Development Set,” Hugging Face dataset, 2026. [Online]. Available: <https://huggingface.co/datasets/Harland/DCASE2026-Task5-DevSet>
- [8] Tongyi Fun Team *et al.*, “Fun-Audio-Chat technical report,” arXiv preprint arXiv:2512.20156, 2025. [Online]. Available: <https://arxiv.org/abs/2512.20156>
- [9] KimiTeam *et al.*, “Kimi-Audio technical report,” arXiv preprint arXiv:2504.18425, 2025. [Online]. Available: <https://arxiv.org/abs/2504.18425>
- [10] Xiaomi LLM-Core Team *et al.*, “MiMo-Audio: Audio language models are few-shot learners,” arXiv preprint arXiv:2512.23808, 2025. [Online]. Available: <https://arxiv.org/abs/2512.23808>
- [11] B. Wu *et al.*, “Step-Audio 2 technical report,” arXiv preprint arXiv:2507.16632, 2025. [Online]. Available: <https://arxiv.org/abs/2507.16632>
- [12] C. Yang *et al.*, “MOSS-Audio technical report,” arXiv preprint arXiv:2606.01802, 2026. [Online]. Available: <https://arxiv.org/abs/2606.01802>
- [13] S.-Y. Liu *et al.*, “GDPO: Group reward-decoupled normalization policy optimization for multi-reward RL optimization,” arXiv preprint arXiv:2601.05242, 2026. [Online]. Available: <https://arxiv.org/abs/2601.05242>
- [14] E. J. Hu, Y. Shen, P. Wallis, Z. Allen-Zhu, Y. Li, S. Wang, L. Wang, and W. Chen, “LoRA: Low-rank adaptation of large language models,” in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2022. [Online]. Available: <https://arxiv.org/abs/2106.09685>
- [15] H. Nam, “Learning from audio-dependency errors: Data curation strategies based on model confusion patterns in audio question answering,” Independent, DCASE2026 Challenge, Jun. 2026.
- [16] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, “QLoRA: Efficient finetuning of quantized LLMs,” in *Proc. NeurIPS*, 2023. [Online]. Available: <https://arxiv.org/abs/2305.14314>
- [17] W. Hu, Y. Cao, and J. Yang, “Audio-dependent question answering at the DCASE 2026 challenge,” Inst. of Acoustics, CAS, DCASE2026 Challenge, Jun. 2026.
- [18] Z. Yin, Q. Cao, P. Deng, Y. Tan, and S. Li, “Training-free inference-time exploration for audio-dependent question answering,” XJTLU, DCASE2026 Challenge, Jun. 2026.
- [19] Y. Cheng, J. Zhao, R. Wan, P. Chang, Y. Chen, and W. Wang, “Selective multi-modal RAG for DCASE 2026 task 5: Audio-dependent question answering,” Univ. of Surrey, DCASE2026 Challenge, Jun. 2026.
- [20] Y. Zhang, J. Liu, and M. Li, “Structured audio reasoning and robust multi-sample inference for DCASE 2026 audio-dependent question answering challenge,” Wuhan Univ. and CUHK-Shenzhen, DCASE2026 Challenge, Jun. 2026.
- [21] A. Tathe, “DCASE 2026 audio-dependent question answering task,” UIUC and CMU, DCASE2026 Challenge, Jun. 2026.
- [22] Q. Huang, Y. Pan, X. Zhou, R. Li, and M. Unoki, “Curriculum learning for audio-dependent question answering: Technical report for DCASE 2026 task 5,” JAIST, DCASE2026 Challenge, Jun. 2026.
- [23] J. Kim, L. Song, and H.-M. Park, “Task-leaf routed MiMo-Audio for DCASE 2026 task 5,” Sogang Univ., DCASE2026 Challenge, Jun. 2026.
- [24] C. Wu, D. Wu, and Y. Huang, “Qwen2.5-Omni with all-audio audio-TAH for audio-dependent question answering,” Xiamen Univ. and Tsinghua Univ., DCASE2026 Challenge, Jun. 2026.
- [25] C. Zheng, “Fun-Audio-Chat-8B with LoRA fine-tuning for audio-dependent question answering,” Hangzhou Dianzi Univ., DCASE2026 Challenge, Jun. 2026.
- [26] F. Xiao, Q. Zhu, and J. Guan, “GISP@HEU’s submission for DCASE 2026 task 5: A LoRA-SFT fine-tuned audio-dependent question answering system,” HEU and UTS, DCASE2026 Challenge, Jun. 2026.
- [27] H. Song, “Audio-dependent question answering with attention-anchored reinforcement learning on MiMo-Audio,” BIT, DCASE2026 Challenge, Jun. 2026.
- [28] H. Xu and R. Zhang, “Audio question answering at the DCASE 2026 challenge,” HUST, DCASE2026 Challenge, Jun. 2026.
- [29] J. Xu *et al.*, “Qwen2.5-Omni technical report,” arXiv preprint arXiv:2503.20215, 2025. [Online]. Available: <https://arxiv.org/abs/2503.20215>
- [30] Z. Shao, P. Wang, Q. Zhu, R. Xu, J. Song, X. Bi, H. Zhang, M. Zhang, Y. K. Li, Y. Wu, and D. Guo, “DeepSeekMath: Pushing the limits of mathematical reasoning in open language models,” arXiv preprint arXiv:2402.03300, 2024. [Online]. Available: <https://arxiv.org/abs/2402.03300>
- [31] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, “Robust speech recognition via large-scale weak supervision,” in *Proc. ICML*, 2023. [Online]. Available: <https://arxiv.org/abs/2212.04356>
- [32] H. Bredin, “pyannote.audio 2.1 speaker diarization pipeline: principle, benchmark, and recipe,” in *Proc. Interspeech*, 2023.
- [33] B. Desplanques, J. Thienpondt, and K. Demuyne, “ECAPA-TDNN: Emphasized channel attention, propagation and aggregation in TDNN based speaker verification,” in *Proc. Interspeech*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.07143>
- [34] Y. Wu, K. Chen, T. Zhang, Y. Hui, M. Nezhurina, T. Berg-Kirkpatrick, and S. Dubnov, “Large-scale Contrastive Language-Audio Pretraining with feature fusion and keyword-to-caption augmentation,” in *Proc. ICASSP*, 2023. [Online]. Available: <https://arxiv.org/abs/2211.06687>
- [35] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, “librosa: Audio and music signal analysis in Python,” in *Proc. SciPy*, 2015.
- [36] DCASE Task 5 Organizers, “ADQA-Bench: Evaluation set,” Hugging Face dataset, 2026. [Online]. Available: <https://huggingface.co/datasets/Harland/ADQA-Bench>
- [37] S. Kim, J. Suk, S. Longpre, B. Y. Lin, J. Shin, S. Welleck, G. Neubig, M. Lee, K. Lee, and M. Seo, “Prometheus 2: An open source language model specialized in evaluating other language models,” arXiv preprint arXiv:2405.01535, 2024. [Online]. Available: <https://arxiv.org/abs/2405.01535>